CS182: Introduction to Machine Learning, Fall 2025
Homework 1 Solution

(Due on Sunday, Oct. 19 at 11:59 pm (CST))

Notice:

- Please submit your assignments via Gradescope. The entry code will be released in a few days.
- Please make sure you select your answer to the corresponding question when submitting your assignments.
- Each person has a total of five days to be late without penalty for all the assignments. Each late delivery less than one day will be counted as one day.

1. [20 points] [*Linear Algebra*] In robotics, the Jacobian Matrix, $\mathbf{J}$, is a fundamental tool that creates a linear mapping between the robot's joint velocities, $\dot{\mathbf{q}}$, and its end-effector velocity, $\mathbf{v}$, through the relationship: $\mathbf{v} = \mathbf{J}\dot{\mathbf{q}}$. By analyzing the properties of the Jacobian, we can evaluate a robot's dexterity and motion capability in a given configuration, an analysis known as manipulability.

   (a) For a robot in a specific configuration, its Jacobian is given by the matrix $\mathbf{J} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$. In manipulability analysis, the symmetric matrix $\mathbf{S} = \mathbf{J}^T\mathbf{J}$ is of critical importance, as its eigenvalues and eigenvectors reveal the intrinsic relationship between joint motions and end-effector motions.

   (a) Calculate the symmetric matrix $\mathbf{S} = \mathbf{J}^T\mathbf{J}$ for this robot.[3 points]

   (b) Find all eigenvalues and their corresponding orthonormal eigenvectors for the matrix $\mathbf{S}$.[3 points]

   (b) The Singular Value Decomposition (SVD) of the Jacobian is given by $\mathbf{J} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. The singular values, σ_i, on the diagonal of $\mathbf{\Sigma}$ are a direct measure of the robot's ability to move in different directions. Prove that the squared singular values of the Jacobian $\mathbf{J}$ are the eigenvalues of the manipulability matrix $\mathbf{J}^T\mathbf{J}$. [6 points]

   (c) The squared magnitude of the end-effector velocity is given by $\|\mathbf{v}\|^2 = \mathbf{v}^T\mathbf{v} = (\mathbf{J}\dot{\mathbf{q}})^T(\mathbf{J}\dot{\mathbf{q}}) = \dot{\mathbf{q}}^T\mathbf{J}^T\mathbf{J}\dot{\mathbf{q}}$. To plan the most efficient movements, engineers need to determine the maximum possible end-effector speed when the joints move with a unit norm speed (i.e., $\|\dot{\mathbf{q}}\|^2 = \dot{\mathbf{q}}^T\dot{\mathbf{q}} = 1$). Prove that for any unit-norm joint velocity vector $\dot{\mathbf{q}}$, the maximum value of the squared end-effector velocity, $\|\mathbf{v}\|^2$, is equal to the largest eigenvalue of the matrix $\mathbf{J}^T\mathbf{J}$.[8 points]

Solution:

(a) Calculate the symmetric matrix $\mathbf{S} = \mathbf{J}^T\mathbf{J}$. First, we find the transpose of the Jacobian matrix $\mathbf{J}$:

$$\mathbf{J} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \implies \mathbf{J}^T = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Next, we compute the product $\mathbf{S} = \mathbf{J}^T\mathbf{J}$:

$$\mathbf{S} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1\cdot 1 + 0\cdot 0 + 1\cdot 1 & 1\cdot 0 + 0\cdot 1 + 1\cdot 1 \\ 0\cdot 1 + 1\cdot 0 + 1\cdot 1 & 0\cdot 0 + 1\cdot 1 + 1\cdot 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Find all eigenvalues and their corresponding orthonormal eigenvectors for $\mathbf{S}$.
We solve the characteristic equation $\det(\mathbf{S} - \lambda\mathbf{I}) = 0$.

$$\begin{aligned} \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} &= 0 \\ (2-\lambda)^2 - 1 &= 0 \\ (2-\lambda-1)(2-\lambda+1) &= 0 \\ (1-\lambda)(3-\lambda) &= 0 \end{aligned}$$

The eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 1$.
For $\lambda_1 = 3$:

$$(\mathbf{S} - 3\mathbf{I})\mathbf{v}_1 = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies -x + y = 0 \implies x = y.$$

An eigenvector is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Normalizing it, we get $\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
For $\lambda_2 = 1$:

$$(\mathbf{S} - 1\mathbf{I})\mathbf{v}_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies x + y = 0 \implies x = -y.$$

An eigenvector is $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Normalizing it, we get $\mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

(b) We start with the definition of the Singular Value Decomposition (SVD) of the Jacobian matrix: $\mathbf{J} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. In this definition, $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices, which means $\mathbf{U}^T\mathbf{U} = \mathbf{I}$ and $\mathbf{V}^T\mathbf{V} = \mathbf{I}$.
We construct the expression for $\mathbf{J}^T\mathbf{J}$:

$$\mathbf{J}^T\mathbf{J} = (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T)^T(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T)$$

Using the property of transposes, $(ABC)^T = C^T B^T A^T$, this becomes:

$$\mathbf{J}^T\mathbf{J} = (\mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^T)(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T)$$

Since $\mathbf{U}$ is an orthogonal matrix, $\mathbf{U}^T\mathbf{U} = \mathbf{I}$. The expression simplifies to:

$$\mathbf{J}^T\mathbf{J} = \mathbf{V}\mathbf{\Sigma}^T(\mathbf{U}^T\mathbf{U})\mathbf{\Sigma}\mathbf{V}^T = \mathbf{V}(\mathbf{\Sigma}^T\mathbf{\Sigma})\mathbf{V}^T$$

Let $\mathbf{D} = \mathbf{\Sigma}^T\mathbf{\Sigma}$. Since $\mathbf{\Sigma}$ is a rectangular diagonal matrix with the singular values σ_i on its diagonal, $\mathbf{D}$ is a square diagonal matrix with the squared singular values σ_i^2 on its diagonal. The equation is now:

$$\mathbf{J}^T\mathbf{J} = \mathbf{V}\mathbf{D}\mathbf{V}^T$$

This is precisely the definition of the eigendecomposition of the matrix $\mathbf{J}^T\mathbf{J}$. This relationship shows that the columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{J}^T\mathbf{J}$, and the diagonal entries of $\mathbf{D}$ are the corresponding eigenvalues.
Therefore, the squared singular values (σ_i^2) of $\mathbf{J}$ are the eigenvalues of $\mathbf{J}^T\mathbf{J}$.

(c) We want to maximize the squared end-effector velocity, $\|\mathbf{v}\|^2 = \dot{\mathbf{q}}^T\mathbf{J}^T\mathbf{J}\dot{\mathbf{q}}$, subject to the constraint that the joint velocity vector has a unit norm, $\|\dot{\mathbf{q}}\|^2 = \dot{\mathbf{q}}^T\dot{\mathbf{q}} = 1$.
Let $\mathbf{S} = \mathbf{J}^T\mathbf{J}$. From Part (a), we know $\mathbf{S}$ is a real, symmetric matrix. Therefore, it possesses a set of orthonormal eigenvectors, $\mathbf{u}_1, \ldots, \mathbf{u}_n$, which form a basis for the vector space. Let their corresponding eigenvalues be ordered such that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$.
Any unit vector $\dot{\mathbf{q}}$ can be expressed as a linear combination of this eigenbasis:

$$\dot{\mathbf{q}} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n$$

The unit norm constraint on $\dot{\mathbf{q}}$ implies that the coefficients must satisfy $\sum_{i=1}^n c_i^2 = 1$, because the eigenvectors are orthonormal.
Now, we substitute the expansion of $\dot{\mathbf{q}}$ into the quadratic form:

$$\dot{\mathbf{q}}^T\mathbf{S}\dot{\mathbf{q}} = \left(\sum_i c_i\mathbf{u}_i^T\right)\mathbf{S}\left(\sum_j c_j\mathbf{u}_j\right)$$

Using the property that $\mathbf{S}\mathbf{u}_j = \lambda_j\mathbf{u}_j$:

$$= \left(\sum_i c_i\mathbf{u}_i^T\right)\left(\sum_j c_j\lambda_j\mathbf{u}_j\right)$$

Due to the orthonormality of the eigenvectors ($\mathbf{u}_i^T\mathbf{u}_j = \delta_{ij}$), the sum simplifies to:

$$\dot{\mathbf{q}}^T\mathbf{S}\dot{\mathbf{q}} = \sum_{i=1}^n c_i^2\lambda_i$$

To maximize this sum, we use the fact that λ_1 is the largest eigenvalue ($\lambda_i \leq \lambda_1$ for all i):

$$\sum c_i^2 \lambda_i \leq \sum c_i^2 \lambda_1 = \lambda_1 \left(\sum c_i^2 \right)$$

Using our constraint that $\sum c_i^2 = 1$:

$$\sum c_i^2 \lambda_i \leq \lambda_1$$

The maximum value of the expression is λ_1. This maximum is achieved when $c_1 = 1$ and all other $c_i = 0$, which corresponds to the case where $\dot{\mathbf{q}}$ is the eigenvector $\mathbf{u}_1$ associated with the largest eigenvalue λ_1.
Thus, the maximum possible squared velocity $\|\mathbf{v}\|^2$ for any unit-norm joint velocity vector is the largest eigenvalue of $\mathbf{J}^T\mathbf{J}$.

Or Our goal is to find the maximum value of the squared end-effector velocity, $\|\mathbf{v}\|^2 = \dot{\mathbf{q}}^T\mathbf{J}^T\mathbf{J}\dot{\mathbf{q}}$, under the constraint that $\|\dot{\mathbf{q}}\|^2 = \dot{\mathbf{q}}^T\dot{\mathbf{q}} = 1$.
Let $\mathbf{S} = \mathbf{J}^T\mathbf{J}$. Since $\mathbf{S}$ is a real, symmetric matrix, it can be diagonalized by an orthogonal matrix of its eigenvectors. Let the eigendecomposition of $\mathbf{S}$ be:

$$\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$$

where $\mathbf{Q}$ is an orthogonal matrix whose columns are the orthonormal eigenvectors of $\mathbf{S}$, and $\mathbf{\Lambda}$ is a diagonal matrix containing the corresponding eigenvalues, λ_i.
Substitute this decomposition into the expression for $\|\mathbf{v}\|^2$:

$$\|\mathbf{v}\|^2 = \dot{\mathbf{q}}^T(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T)\dot{\mathbf{q}}$$

By regrouping the terms, we get:

$$\|\mathbf{v}\|^2 = (\dot{\mathbf{q}}^T\mathbf{Q})\mathbf{\Lambda}(\mathbf{Q}^T\dot{\mathbf{q}})$$

Let us define a new vector, $\mathbf{a}$, which represents the coordinates of $\dot{\mathbf{q}}$ in the basis of eigenvectors:

$$\mathbf{a} = \mathbf{Q}^T\dot{\mathbf{q}}$$

Then, its transpose is $\mathbf{a}^T = (\mathbf{Q}^T\dot{\mathbf{q}})^T = \dot{\mathbf{q}}^T\mathbf{Q}$. The expression for $\|\mathbf{v}\|^2$ simplifies to:

$$\|\mathbf{v}\|^2 = \mathbf{a}^T\mathbf{\Lambda}\mathbf{a}$$

Since $\mathbf{\Lambda}$ is a diagonal matrix of the eigenvalues λ_i, this quadratic form expands to a simple weighted sum:

$$\|\mathbf{v}\|^2 = \begin{pmatrix} a_1 & \dots & a_n \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \sum_{i=1}^{n} a_i^2 \lambda_i$$

Next, we examine how the constraint on $\dot{\mathbf{q}}$ transforms for the new vector $\mathbf{a}$.

$$\|\mathbf{a}\|^2 = \mathbf{a}^T\mathbf{a} = (\mathbf{Q}^T\dot{\mathbf{q}})^T(\mathbf{Q}^T\dot{\mathbf{q}}) = \dot{\mathbf{q}}^T\mathbf{Q}\mathbf{Q}^T\dot{\mathbf{q}}$$

Because $\mathbf{Q}$ is an orthogonal matrix, $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$ (the identity matrix). Therefore:

$$\|\mathbf{a}\|^2 = \dot{\mathbf{q}}^T\mathbf{I}\dot{\mathbf{q}} = \dot{\mathbf{q}}^T\dot{\mathbf{q}} = \|\dot{\mathbf{q}}\|^2 = 1$$

This means that $\mathbf{a}$ is also a unit vector, and its components satisfy the condition $\sum_{i=1}^{n} a_i^2 = 1$.
Our problem is now reduced to maximizing the sum $\sum a_i^2\lambda_i$ subject to the constraint $\sum a_i^2 = 1$. Let λ_{max} be the largest eigenvalue among all λ_i. We can then write:

$$\sum a_i^2 \lambda_i \leq \sum a_i^2 \lambda_{max} = \lambda_{max} \left(\sum a_i^2 \right)$$

Since $\sum a_i^2 = 1$, the inequality becomes:

$$\sum a_i^2 \lambda_i \leq \lambda_{max}$$

The maximum value of the expression is λ_{max}, which is achieved when the component a_k corresponding to λ_{max} is 1 (and all other components are 0). This corresponds to the case where $\dot{\mathbf{q}}$ is the eigenvector associated with the largest eigenvalue.
Thus, the maximum value of $\|\mathbf{v}\|^2$ is the largest eigenvalue of $\mathbf{J}^T\mathbf{J}$.

2. [20 points] [*Linear Algebra*] Let $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T A\mathbf{x} - \boldsymbol{b}^T\mathbf{x} + c$, where $A \in \mathbb{R}^{n\times n}$ is symmetric, $\boldsymbol{b} \in \mathbb{R}^n$, and $c \in \mathbb{R}$.

   (a) Derive the Gradient and Hessian of $f(\mathbf{x})$. [8 points]

   (b) Under what condition on A does $f(\mathbf{x})$ correspond to a minimum, and find the point $\mathbf{x}^*$ that makes $f(\mathbf{x})$ smallest. [6 points]

   (c) Suppose A has eigenvalues λ_1, λ_2. Describe the general shape of $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T A\mathbf{x}$ in 2D for the following cases.

   (1) $\lambda_1, \lambda_2 > 0$. [2 points]
   (2) $\lambda_1, \lambda_2 < 0$. [2 points]
   (3) $\lambda_1 > 0, \lambda_2 < 0$. [2 points]

Solution:

(a) We use standard matrix calculus results:

$$\nabla_{\mathbf{x}}(\mathbf{x}^T A\mathbf{x}) = (A + A^T)\mathbf{x}, \quad \text{and since } A = A^T, \text{ we have } \nabla_{\mathbf{x}}(\mathbf{x}^T A\mathbf{x}) = 2A\mathbf{x}.$$

Then,

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \frac{1}{2}(2A\mathbf{x}) - \boldsymbol{b} = A\mathbf{x} - b.$$

For the second derivative,

$$\nabla^2_{\mathbf{x}} f(\mathbf{x}) = \nabla_{\mathbf{x}}(A\mathbf{x} - \boldsymbol{b}) = A,$$

since A is constant with respect to $\mathbf{x}$.

(b) The critical point ($\mathbf{x}^*$) satisfies

$$\nabla_{\mathbf{x}} f(\mathbf{x}^*) = 0 \quad \Longrightarrow \quad A\mathbf{x}^* - \boldsymbol{b} = 0.$$

Thus,

$$\mathbf{x}^* = A^{-1}\boldsymbol{b}.$$

To determine whether this point is a minimum, we check the definiteness of the Hessian:

$$\nabla^2_{\mathbf{x}} f(\mathbf{x}) = A$$

If A is positive definite ($\mathbf{x}^T A\mathbf{x} > 0$ for all $\mathbf{x} \neq 0$), then $f(\mathbf{x})$ is strictly convex, and $\mathbf{x}^*$ is the unique global minimizer.
If A is positive semidefinite ($\mathbf{x}^T A\mathbf{x} \geq 0$ for all $\mathbf{x} \neq 0$), $f(\mathbf{x})$ is convex but may have multiple minima.

(c) Because A is symmetric, it can be diagonalized as $A = Q\Lambda Q^T$, where $\Lambda = \text{diag}(\lambda_1, \lambda_2)$.
Let $\mathbf{z} = Q^T\mathbf{x}$, the quadratic form can be rewritten as

$$f(\mathbf{x}) = \frac{1}{2}(\lambda_1 z_1^2 + \lambda_2 z_2^2),$$

When $\lambda_1, \lambda_2 > 0$, the surface curves upward in every direction. The graph of $f(\mathbf{x})$ is a bowl-shaped paraboloid, which is convex. The function has a unique minimum at the origin.
When $\lambda_1, \lambda_2 > 0$, the surface curves downward in every direction. The graph of $f(\mathbf{x})$ is an inverted bowl, which is concave. The function has a maximum at the origin.
When $\lambda_1, \lambda_2 > 0$, the surface curves upward in one direction and downward in the other. The graph of $f(\mathbf{x})$ forms a saddle-shaped surface, and the origin is a saddle point rather than a minimum or a maximum.

3. [20 points] [*Probability and Statistics*]

(a) For any non-negative random variable X with finite variance, prove that

$$\Pr[X > 0] \geq \frac{(\mathrm{E}[X])^2}{\mathrm{E}[X^2]}.$$

Hint: Just use the fact that the expectation $\mathrm{E}[UV]$ is a valid inner product of random variables U and V and you can therefore use the Cauchy-Schwarz inequality. [4 points]

(b) For a random variable X with finite variance and E[X] = 0, prove that

$$\Pr[X \geq t] \leq \frac{\mathrm{E}[X^2]}{\mathrm{E}[X^2] + t^2} \text{ for any } t \geq 0$$

Hint: You can try the same logic as in the previous question.[6 points]

(c) Prove that the following definition of Gaussian random vector is equivalent to the definition via the density function:

Definition 1 *Gaussian random vectors are affine transformations of IID gaussian random variables. In other words, if X has nonsingular $\mathbf{\Sigma_X}$, then we can write*

$$\mathbf{X} = \mu_{\mathbf{x}} + \mathbf{AW}$$

for some full rank $\mathbf{A} \in \mathbb{R}^{n \times n}$ and $\mathbf{W} = (W_1, ..., W_n)^T$, where $W_i \sim_{\text{IID}} \mathcal{N}(0, 1)$

Hint: For strictly monotone and differentiable function $g(x)$ such that $Y = g(X)$, and let g^{-1} be the inverse function of g

$$f_Y(y) = f_X(g^{-1}(y)) \frac{1}{|g'(g^{-1}(y)|}$$

[10 points]

Solution:

(a) With the non-negativity of X, we have $\mathbb{E}[X] = \mathbb{E}[X \cdot \mathbf{1}\{X > 0\}]$.
Next we apply Cauchy-Schwarz to $U = X$ and $V = \mathbf{1}\{X > 0\}$ and conclude that

$$(\mathbb{E}[X])^2 = (\mathbb{E}[X\mathbf{1}\{X > 0\}])^2 \leq \mathbb{E}[X^2]\mathbb{E}[\mathbf{1}\{X > 0\}^2] = \mathbb{E}[X^2]\mathbb{E}[\mathbf{1}\{X > 0\}] = \mathbb{E}[X^2]\mathbb{P}(X > 0).$$

(b) Similar to (a),

$$\mathbb{E}[t - X] \leq \mathbb{E}[(t - X)\mathbf{1}\{t - X > 0\}] = \mathbb{E}[(t - X)\mathbf{1}\{X < t\}].$$

Now apply Cauchy-Schwarz to get

$$(\mathbb{E}[t - X])^2 \leq (\mathbb{E}[(t - X)\mathbf{1}(X < t)])^2 \leq \mathbb{E}[(t - X)^2]\mathbb{E}[\mathbf{1}(X < t)].$$

Evaluate the terms on the right-hand side and left-hand side separately.
Since $\mathbb{E}[X] = 0$, the LHS is

$$(\mathbb{E}[t - X])^2 = t^2$$

and the RHS becomes

$$\mathbb{E}[(t - X)^2]\mathbb{E}[\mathbf{1}(X < t)] = (t^2 + \mathbb{E}[X^2])(1 - \mathbb{P}(X \geq t))$$

After some rearranging we obtain

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X^2]}{\mathbb{E}[X^2] + t^2}$$

as desired.

(c) **definition 1 $\Rightarrow$ definition via PDF**
For strictly monotone(say increasing) and differentiable function $g(x)$ such that $Y = g(X)$, with inverse g^{-1}

$$\begin{aligned} f_Y(y) = f_X(g^{-1}(y)) &= \frac{\mathrm{d}}{\mathrm{d}y} \Pr[g(X) \leq y] \\ &= \frac{\mathrm{d}}{\mathrm{d}y} \Pr[X \leq g^{-1}(y)] \\ &= f_X(g^{-1}(y)) \frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y) \\ &= f_X(g^{-1}(y)) \frac{1}{g'(g^{-1}(y))} \end{aligned}$$

When g is strictly monotone decreasing, the only change is a minus sign in front of $g'(g^{-1}(y))$. Thus, the two cases can be summarized in the single formula:

$$f_Y(y) = f_X(g^{-1}(y))\frac{1}{|g'(g^{-1}(y)|} \tag{1}$$

Let X be the random vectors in definition 1

$$\begin{aligned} f_X(x) &= \frac{1}{|\det(A)|} f_W(A^{-1}(x-\mu_X)) \\ &= \frac{1}{|\det(A)|}\frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}|A^{-1}(x-\mu_X)|^2} \\ &= \frac{1}{|\det(AA^T)|^{\frac{1}{2}}(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}(x-\mu_X)^T(AA^T)^{-1}(x-\mu_X)} \end{aligned}$$

We also know that

$$\begin{aligned} \Sigma_X &= E\left[(X-E[X])(X-E[X])^T\right] \\ &= E\left[(X-\mu_X)(X-\mu_X)^T\right] \\ &= E\left[AWW^TA^T\right] \\ &= AIA^T \\ &= AA^T \end{aligned}$$

definition via PDF $\Rightarrow$ definition 1
For Gaussian random vector defined by the probability density function, set $\mu_x = E[X]$ and $A = \Sigma_x^{1/2}$, we can easily obtain

$$X = \mu_x + AW$$

where $W \sim \mathcal{N}(0, I)$, which completes the proof.

4. [20 points] [*Bayesian Decision Theory*] A major technology company is hiring for a Machine Learning Engineer position and uses a two-stage screening process:

   - 1. Resume screening: (binary outcome: *Pass* or *Fail*)
   - 2. Technical interview (binary outcome: *Hire* or *Reject*)

   Historical hiring data reveals the following:

   - 20% of all applicants are ultimately qualified for the role. (i.e., $P(\omega_Q) = 0.2$, where ω_Q = qualified, ω_U = unqualified).
   - Among qualified candidates, 90% pass the resume screen: $P(\text{Pass} \mid \omega_Q) = 0.9$.
   - Among unqualified candidates, 30% pass the resume screen: $P(\text{Pass} \mid \omega_U) = 0.3$.
   - The company incurs different costs for hiring decisions:
     - Correctly hiring a qualified candidate yields a benefit of +10 (i.e., cost = –10).
     - Rejecting a qualified candidate has a cost of 5 (missed opportunity)
     - Hiring an unqualified candidate has a cost of 15 (training, turnover, etc.).
     - Correctly rejecting an unqualified candidate has zero cost.

   (a) Compute the posterior probability that an applicant who passed the resume screen is actually qualified, i.e. $P(\omega_U \mid \text{Pass})$.[4 points]

   (b) Using Bayesian decision theory, determine whether the company should automatically hire every applicant who passes the resume screen, or reject them, based on minimizing expected risk. Justify your answer by computing the conditional risk for both actions.[4 points]

   (c) Suppose the company introduces a new AI-powered resume screener that improves the true positive rate to 95% while reducing the false positive rate to 10%. Recalculate the posterior $P(\omega_Q \mid \text{Pass})$ and the optimal decision in question (b). Discuss how advancements in screening technology affect hiring decisions from a Bayesian perspective. [8 points]

   (d) Discuss how advancements in screening technology affect hiring decisions from a Bayesian perspective.[4 points]

   Solution:

   We are given the following prior probabilities:

$$P(\omega_Q) = 0.2 \, P(\omega_U) = 1 - P(\omega_Q) = 0.8,$$

   and likelihoods for resume screening:

$$P(\text{Pass} \mid \omega_Q) = 0.9 \, P(\text{Pass} \mid \omega_U) = 0.3.$$

   (a) Using Bayesian formula, one has:

$$\begin{aligned} P(\omega_Q \mid \text{Pass}) =& \frac{P(\text{Pass} \mid \omega_Q) P(\omega_Q)}{P(\text{Pass})} \\ =& \frac{P(\text{Pass} \mid \omega_Q) P(\omega_Q)}{P(\text{Pass} \mid \omega_Q) P(\omega_Q) + P(\text{Pass} \mid \omega_U) P(\omega_U)} \\ =& \frac{0.9 * 0.2}{0.9 * 0.2 + 0.3 * 0.8} \\ =& \frac{3}{7} \\ \approx& 42.86\%. \end{aligned}$$

   Therefore, about 42.86% of applicants who pass the resume screen are actually qualified.

   (b) We compute the conditional risk $R(\alpha_i \mid \text{Pass})$ for each action, where α_1 = Hire and α_2 = Reject.

   - For $R(\alpha_1 \mid \text{Pass})$, one has

$$\begin{aligned} R(\alpha_1 \mid \text{Pass}) =& \lambda(\alpha_1 \mid \omega_Q) P(\omega_Q \mid \text{Pass}) + \lambda(\alpha_1 \mid \omega_U) P(\omega_U \mid \text{Pass}) \\ =& -10 * \frac{3}{7} + 15 * \frac{4}{7} \\ =& \frac{30}{7}. \end{aligned}$$

- For $R(\alpha_2 \mid \text{Pass})$, one has

$$\begin{aligned} R(\alpha_2 \mid \text{Pass}) =& \lambda(\alpha_2 \mid \omega_Q) P(\omega_Q \mid \text{Pass}) + \lambda(\alpha_2 \mid \omega_U) P(\omega_U \mid \text{Pass}) \\ =& 5 * \frac{3}{7} + 0 * \frac{4}{7} \\ =& \frac{15}{7}. \end{aligned}$$

Owing to $R(\alpha_2 \mid \text{Pass}) < R(\alpha_1 \mid \text{Pass})$, the optimal decision is to reject applicants who only pass the resume screen.

(c) We have the new prior information:

$$P(\text{Pass} \mid \omega_Q) = 0.95\, P(\text{Pass} \mid \omega_U) = 0.1,$$

which leads to

$$\begin{aligned} P(\omega_Q \mid \text{Pass}) =& \frac{P(\text{Pass} \mid \omega_Q) P(\omega_Q)}{P(\text{Pass})} \\ =& \frac{P(\text{Pass} \mid \omega_Q) P(\omega_Q)}{P(\text{Pass} \mid \omega_Q) P(\omega_Q) + P(\text{Pass} \mid \omega_U) P(\omega_U)} \\ =& \frac{0.95 * 0.2}{0.95 * 0.2 + 0.1 * 0.8} \\ =& \frac{19}{27} \\ \approx& 70.37\%. \end{aligned}$$

Therefore, about 70.37% of applicants who pass the resume screen are actually qualified. We than recompute $R(\alpha_i \mid \text{Pass})$, where

$$\begin{aligned} R(\alpha_1 \mid \text{Pass}) =& \lambda(\alpha_1 \mid \omega_Q) P(\omega_Q \mid \text{Pass}) + \lambda(\alpha_1 \mid \omega_U) P(\omega_U \mid \text{Pass}) \\ =& -10 * \frac{19}{27} + 15 * \frac{8}{27} \\ =& \frac{-70}{27}, \end{aligned}$$

and

$$\begin{aligned} R(\alpha_2 \mid \text{Pass}) =& \lambda(\alpha_2 \mid \omega_Q) P(\omega_Q \mid \text{Pass}) + \lambda(\alpha_2 \mid \omega_U) P(\omega_U \mid \text{Pass}) \\ =& 5 * \frac{19}{27} + 0 * \frac{8}{27} \\ =& \frac{95}{27}. \end{aligned}$$

Since $R(\alpha_1 \mid \text{Pass}) < R(\alpha_2 \mid \text{Pass})$, now we can choose to hire all applicants.

- For $R(\alpha_1 \mid \text{Pass})$, one has

$$\begin{aligned} R(\alpha_1 \mid \text{Pass}) =& \lambda(\alpha_1 \mid \omega_Q) P(\omega_Q \mid \text{Pass}) + \lambda(\alpha_1 \mid \omega_U) P(\omega_U \mid \text{Pass}) \\ =& -10 * \frac{3}{7} + 15 * \frac{4}{7} \\ =& \frac{30}{7}. \end{aligned}$$

- For $R(\alpha_2 \mid \text{Pass})$, one has

$$\begin{aligned} R(\alpha_2 \mid \text{Pass}) =& \lambda(\alpha_2 \mid \omega_Q) P(\omega_Q \mid \text{Pass}) + \lambda(\alpha_2 \mid \omega_U) P(\omega_U \mid \text{Pass}) \\ =& 5 * \frac{3}{7} + 0 * \frac{4}{7} \\ =& \frac{15}{7}. \end{aligned}$$

Owing to $R(\alpha_2 \mid \text{Pass}) < R(\alpha_1 \mid \text{Pass})$, the optimal decision is to reject applicants who only pass the resume screen.

(d) In Bayesian decision theory, an optimal decision is made by minimizing expected risk, which depends on two key components in our question:

- The posterior probabilities: $P(\omega_j \mid \boldsymbol{x})$—our updated belief about a candidate's true state (qualified/unqualified) after observing evidence $\boldsymbol{x}$ (e.g., "passed resume screen")
- Loss function: $\lambda(\alpha_i \mid \omega_j)$, the cost of taking action α_i (e.g., hire/reject) when the true state is ω_j.

Advances in screening technology, such as AI-powered resume parsers, coding assessment platforms, or behavioral signal analysis—directly improve the likelihood model $P(\boldsymbol{x} \mid \omega_j)$. Specificaly, they may increase the true positive rate: $P(\text{Pass} \mid \omega_Q)$ and decrease the false positive rate $P(\text{Pass} \mid \omega_U)$. By Baye's theorem:

$$P(\omega_Q \mid \text{Pass}) = \frac{P(\text{Pass} \mid \omega_Q) P(\omega_Q)}{P(\text{Pass} \mid \omega_Q) P(\omega_Q) + P(\text{Pass} \mid \omega_U) P(\omega_U)} \quad ,$$

the posterior probability that a candidate who passes the screen is truly qualified can be increased by improving the likelihood ratio $P(\text{Pass} \mid \omega_Q) / P(\text{Pass} \mid \omega_U)$.

Furthermore, owing to $\lambda(\alpha_1 \mid \omega_Q)$ is negative , the conditional risk

$$R(\alpha_1 \mid \text{Pass}) = \lambda(\alpha_1 \mid \omega_Q) P(\omega_Q \mid \text{Pass}) + \lambda(\alpha_1 \mid \omega_U) P(\omega_U \mid \text{Pass})$$

may decrease, which leads to the optimal decision to tend toward full employment.

5. [20 points] [*Bayesian Decision Theory*] Given a training dataset $T = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)\}$, where $\mathbf{x}_i = (x_i^{(1)}, x_i^{(2)}, \ldots, x_i^{(n)})$ is a feature vector and $y_i \in \{c_1, c_2, \ldots, c_K\}$ is a class label. Bayesian estimation often add a small constant λ to each count, i.e.,

$$P_\lambda(Y = c_k) = \frac{\sum_{i=1}^{N} I(y_i = c_k) + \lambda}{N + K\lambda}, \quad P_\lambda(X^{(j)} = a_{jl}|Y = c_k) = \frac{\sum_{i=1}^{N} I(x_i^{(j)} = a_{jl}, y_i = c_k) + \lambda}{\sum_{i=1}^{N} I(y_i = c_k) + S_j\lambda}.$$

where S_j is the number of possible values for feature j.

Table 1: Dataset for playing game

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Rainy	Hot	Yes
Rainy	Mild	Yes

Consider the dataset T in Table 1 which records whether a game is played based on the outlook and temperature. The features are: Outlook: Sunny (S) or Rainy (R), Temperature (Temp): Hot (H) or Mild (M). The class label Play include: Yes (Y) or No (N).

(a) Estimate the prior probabilities $P(\text{Play})$ and the conditional probabilities $P(\text{Outlook}|\text{Play})$ and $P(\text{Temp}|\text{Play})$ using maximum likelihood estimation. Then, for a new instance with Outlook=Rainy and Temp=H, compute the posterior probabilities $P(\text{Play} = \text{Y}|\text{Outlook} = \text{R}, \text{Temp} = \text{H})$ and $P(\text{Play} = \text{N}|\text{Outlook} = \text{R}, \text{Temp} = \text{H})$ using the naive Bayes classifier. [10 points]

(b) Use Bayesian estimation with Laplacian smoothing (with smoothing parameter $\lambda = 1$). Recompute the prior probabilities (with smoothing) and the conditional probabilities. Then, compute the posterior probabilities for the same new instance. [5 points]

(c) Based on the results of the above calculations, discuss why Laplacian smoothing is used. [5 points]

Solution:

(a) First, compute the prior probabilities:

$$P(\text{Play} = \text{Y}) = \frac{2}{4} = 0.5, \quad P(\text{Play} = \text{N}) = \frac{2}{4} = 0.5$$

Now, compute the conditional probabilities for Outlook given Play:

$$P(\text{Outlook} = \text{S}|\text{Play} = \text{Y}) = \frac{0}{2} = 0$$

$$P(\text{Outlook} = \text{R}|\text{Play} = \text{Y}) = \frac{2}{2} = 1$$

$$P(\text{Outlook} = \text{S}|\text{Play} = \text{N}) = \frac{2}{2} = 1$$

$$P(\text{Outlook} = \text{R}|\text{Play} = \text{N}) = \frac{0}{2} = 0$$

Similarly, for Temperature given Play:

$$P(\text{Temp} = \text{H}|\text{Play} = \text{Y}) = \frac{1}{2} = 0.5$$

$$P(\text{Temp} = \text{M}|\text{Play} = \text{Y}) = \frac{1}{2} = 0.5$$

$$P(\text{Temp} = \text{H}|\text{Play} = \text{N}) = \frac{1}{2} = 0.5$$

$$P(\text{Temp} = \text{M}|\text{Play} = \text{N}) = \frac{1}{2} = 0.5$$

For the new instance (Outlook=R, Temperature=H), compute the posterior probabilities using naive Bayes:

$$\begin{aligned}&P(\text{Play} = \text{Y}|\text{Outlook} = \text{R}, \text{Temp} = \text{H})\\&\propto P(\text{Play} = \text{Y}) \cdot P(\text{Outlook} = \text{R}|\text{Play} = \text{Y}) \cdot P(\text{Temp} = \text{H}|\text{Play} = \text{Y})\\&= 0.5 \cdot 1 \cdot 0.5 = 0.25\end{aligned}$$

and

$$\begin{aligned}&P(\text{Play} = \text{N}|\text{Outlook} = \text{R}, \text{Temp} = \text{H})\\&\propto P(\text{Play} = \text{N}) \cdot P(\text{Outlook} = \text{R}|\text{Play} = \text{N}) \cdot P(\text{Temp} = \text{H}|\text{Play} = \text{N})\\&= 0.5 \cdot 0 \cdot 0.5 = 0\end{aligned}$$

After normalization, $P(\text{Yes}|R, H) = 1$ and $P(\text{No}|R, H) = 0$.

(b) With Laplacian smoothing, we add $\lambda = 1$ to each count and adjust the denominator. For Outlook, the number of possible values is 2 (Sunny and Rainy). So, for $P(\text{Outlook}|\text{Play})$:
For Play = Y:

$$\begin{aligned}P(\text{Play} = \text{Y}) &= \frac{2+1}{4+2} = \frac{1}{2} = 0.5,\\P(\text{Outlook} = \text{S}|\text{Play} = \text{Y}) &= \frac{0+1}{2+2} = \frac{1}{4} = 0.25,\\P(\text{Outlook} = \text{R}|\text{Play} = \text{Y}) &= \frac{2+1}{2+2} = \frac{3}{4} = 0.75.\end{aligned}$$

For Play = N:

$$\begin{aligned}P(\text{Play} = \text{N}) &= \frac{2+1}{4+2} = \frac{1}{2} = 0.5,\\P(\text{Outlook} = \text{S}|\text{Play} = \text{N}) &= \frac{2+1}{2+2} = \frac{3}{4} = 0.75,\\P(\text{Outlook} = \text{R}|\text{Play} = \text{N}) &= \frac{0+1}{2+2} = \frac{1}{4} = 0.25.\end{aligned}$$

For Temperature, number of possible values is 2 (Hot and Mild). For Play=Yes:

$$\begin{aligned}P(\text{Temp} = \text{H}|\text{Play} = \text{Y}) &= \frac{1+1}{2+2} = \frac{2}{4} = 0.5\\P(\text{Temp} = \text{M}|\text{Play} = \text{Y}) &= \frac{1+1}{2+2} = \frac{2}{4} = 0.5\end{aligned}$$

For Play=No:

$$\begin{aligned}P(\text{Temp} = \text{H}|\text{Play} = \text{N}) &= \frac{1+1}{2+2} = \frac{2}{4} = 0.5\\P(\text{Temp} = \text{M}|\text{Play} = \text{N}) &= \frac{1+1}{2+2} = \frac{2}{4} = 0.5\end{aligned}$$

Now, for the new instance (Outlook=R, Temperature=H):

$$\begin{aligned}&P(\text{Play} = \text{Y}|\text{Outlook} = \text{R}, \text{Temp} = \text{H})\\&\propto P(\text{Play} = \text{Y}) \cdot P(\text{Outlook} = \text{R}|\text{Play} = \text{Y}) \cdot P(\text{Temp} = \text{H}|\text{Play} = \text{Y})\\&= 0.5 \cdot 0.75 \cdot 0.5 = 0.1875\end{aligned}$$

and

$$\begin{aligned}&P(\text{Play} = \text{N}|\text{Outlook} = \text{R}, \text{Temp} = \text{H})\\&\propto P(\text{Play} = \text{N}) \cdot P(\text{Outlook} = \text{R}|\text{Play} = \text{N}) \cdot P(\text{Temp} = \text{H}|\text{Play} = \text{N})\\&= 0.5 \cdot 0.25 \cdot 0.5 = 0.0625\end{aligned}$$

After normalization:

$$\begin{aligned}P(\text{Play} = \text{Y}|\text{Outlook} = \text{R}, \text{Temp} = \text{H}) &= \frac{0.1875}{0.1875 + 0.0625} = \frac{0.1875}{0.25} = 0.75,\\P(\text{Play} = \text{N}|\text{Outlook} = \text{R}, \text{Temp} = \text{H}) &= \frac{0.0625}{0.25} = 0.25.\end{aligned}$$

(c) In (a), the zero probability occurs because $P(\text{Outlook} = \text{R}|\text{Play} = \text{N}) = 0$, which makes the posterior probability zero for Play=No. This is problematic because it overfits the training data and may not generalize well. With smoothing, the zero probability is avoided, and both classes have non-zero probabilities. Bayesian estimation with smoothing avoids the zero-probability problem, thereby enhancing the generalization ability of the model.